

Department of Computer Science
Computer Networks
CSE 421, SEC 3,4,16
QUIZ 2,SET A
Time 25 min, Marks: 15

ID	Name:	Sec
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1.	A. An ecommerce company called “e-buys” has its own DNS server. The Resource Records (RR) for this company in the top level server “.com” are : (e-buys.com, dnscontrol.e-buys.com, NS) and (dnscontrol.e-buys.com, 200.112.10.1, A). Their web server “e-buys” has the IP address 200.112.30.5. First state the RR for the web server that they need to put in their own DNS server. B. You enter “www.e-buys.com” in your PC browser to visit their website. Draw or list the steps required to find the IP address, without putting extra burden on the Root Servers.	5+ 5= 10
2.	You visit www.daraz.com using Google Chrome browser at 9:30 AM. Later, at 6:00 PM, you revisit the website using the Mozilla Firefox browser. According to the concept of cookies, you should have been able to view your previous searches when you visit the same website. But in this scenario, you did not see anything related to the last time you visited the website. Identify the reason behind this.	5

1. RR to Add in e-buys's Own DNS Server

The company e-buys must add a Resource Record (RR) in its DNS server for its web server:(www.e-buys.com, 200.112.30.5, A). This maps the hostname **www.e-buys.com** to the IP address **200.112.30.5**.

2. Cookies are browser-specific. This means:

- When you visited **www.daraz.com** using **Google Chrome** at 9:30 AM, the website may have stored session data (like search history) in a **cookie** in Chrome.
- When you revisited the website at 6:00 PM using **Mozilla Firefox**, that browser did **not have access** to the cookies stored by Chrome.

Conclusion: Browsers **do not share cookies**. Each maintains its own isolated cookie store for privacy and security reasons. That's why the site couldn't show your previous activity.

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QUIZ 2,SET B
Time 25 min, Marks: 15

ID	Name:	Sec
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1.	A. Bob uses Chrome to open his email account and has sent Charlie an email. Charlie receives the email in such a manner that he is able to read it only once. Choose the correct protocol/s involved between and discuss briefly I. Bob's PC and Bob's Email Server (state in one line why this protocol was used) II. Bob's Email server and Charlie's email server III. Charlie's email server and Charlie's PC. B. Explain which protocol is better, IMAP or POP3, when using your mobile phone for emails?	5+ 5= 10
2.	Monte visited www.ahar.com on 24th January 2023 at 10:30 AM with a DNS TTL of 5 hours. Later, at 4:00 PM, Phonte visited the same website with a DNS TTL of 10 hours and was able to access it. However, when Monte tried to revisit the website at 4:30 PM, he was unable to see the site. What could be the possible reason for Monte's inability to access www.ahar.com?	5

1. Smtip
2. Smtip
3. pop3/imap

B. IMAP is better for mobile use.

Why IMAP?

- Syncs across devices: You can read, delete, and manage emails from any device (PC, phone, tablet).
Emails stay on the server.
More flexible and suitable for people using multiple devices.
Why POP3 is not ideal:
- Emails are typically **downloaded and removed** from the server.
Not synced across devices — if you check email on one device, it may not appear on another.

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QUIZ 2,SET C
Time 25 min, Marks: 15

ID	Name:	Sec
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1.	Given, you want to visit “welcomebacktophysicalexams.com”. Your network has a local DNS server and is using a recursive lookup to fetch IP addresses and RTT of 29ms. Others in your network have already visited the above website a few hours before your visit. After fetching the IP address, your PC sends a persistent HTTP request of size 109 bytes to bring 13 objects of size 31 bytes each. In this particular connection, it requires the PC 67ms to send the TCP request to the server. I. What is DNS lookup and what is the total RTT in ms in the DNS lookup? II. What is the total RTT in ms required (including DNS) to bring the full object? III. What is the total file transmission time in ms after fetching the IP address?	5x 3= 15
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